

ECEN-629: Homework 1

Due on September 16, 2025 at 12:45 pm

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Problem 1

Compute the norm of the following vector $x = (-100, -99, -98, \dots, 99, 100)^T$.

- (a) ℓ_1 norm;
- (b) ℓ_∞ norm;
- (c) ℓ_2 norm.

Solution

We have the definition of a norm as

$$\|x\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}$$

Then for each part we have:

$$\begin{aligned} \ell_1 : \|x\|_1 &= |-100| + |-99| + \dots + |99| + |100| \\ &= 2(1 + 2 + \dots + 99 + 100) \\ &= 2 \cdot \frac{100 \cdot 101}{2} \\ &= 10100 \end{aligned}$$

$$\begin{aligned} \ell_\infty : \|x\|_\infty &= \max\{|x|\} \\ &= 100 \end{aligned}$$

$$\begin{aligned} \ell_2 : \|x\|_2 &= \sqrt{(-100)^2 + (-99)^2 + \dots + (99)^2 + (100)^2} \\ &= \sqrt{2(1^2 + 2^2 + \dots + 99^2 + 100^2)} \\ &= \sqrt{2 \cdot \frac{100 \cdot 101 \cdot 201}{6}} \\ &= \sqrt{676700} \\ &\approx 822.62 \end{aligned}$$

Problem 2

For a quadratic function $f(x) = x^T P x + q^T x + r$ where P is not necessarily symmetric, show that the gradient is $\nabla f(x) = (P + P^T)x + q$. Find the Hessian of the function. Provide details in your proofs.

Solution

I start by decomposing $f(x)$ into summations:

$$f(x) = \sum_{i=1}^m \sum_{j=1}^m P_{ij} x_i x_j + \sum_{i=1}^m q_i x_i + r$$

Then we find the partials of the first term:

$$\begin{aligned} \frac{\partial}{\partial x_k} f(x) &= \frac{\partial}{\partial x_k} \left(\sum_{i=1}^m \sum_{j=1}^m P_{ij} x_i x_j \right) = \sum_{i=1}^m \sum_{j=1}^m \frac{\partial}{\partial x_k} (P_{ij} x_i x_j) \\ &= \sum_{i=1}^m P_{kj} x_j + \sum_{j=1}^m P_{ik} x_i \end{aligned}$$

And the second term:

$$\frac{\partial}{\partial x_k} \left(\sum_{i=1}^m q_i x_i + r \right) = q_k$$

The final term r is constant, so its derivative is zero.

Now we can combine these indexed derivatives to get the gradient:

$$\nabla f(x) = (P + P^T)x + q$$

Now the Hessian of $f(x)$ is found by taking the partials of the gradient we just found:

$$\mathbf{H} = \nabla^2 f(x) = P + P^T$$

Problem 3

A matrix $A \in \mathbb{R}^{m \times n}$ can also be viewed as a vector, by stacking all its n columns into a single length- mn column. Let $x \in \mathbb{R}^m$, $y \in \mathbb{R}^n$ be two column vectors. Show that

$$\text{tr}(A^T xy^T) = \langle xy^T, A \rangle = x^T Ay,$$

where tr is the trace of the matrix.

Solution

We know that rotation of our trace elements gives us

$$\text{tr}(A^T xy^T) = \text{tr}(y^T A^T x)$$

Here the product inside the trace is a 1×1 matrix, thus

$$\text{tr}(y^T A^T x) = y^T A^T x = x^T Ay$$

The inner product is defined as

$$\langle xy^T, A \rangle = \sum_{i=1}^m \sum_{j=1}^n (xy^T)_{ij} A_{ij} = \sum_{i=1}^m \sum_{j=1}^n x_i y_j A_{ij} = x^T Ay$$

Thus both the trace of $A^T xy^T$ and the inner product $\langle xy^T, A \rangle$ equal $x^T Ay$.

Problem 4

Let $B \in \mathbb{R}^{n \times n}$ be a given symmetric matrix of size n -by- n . Determine if the following set is convex: the set of all n -by- n symmetric A matrices, such that $A - B$ is positive semidefinite. If it is convex, provide your reasoning; if not, provide a counter-example.

Solution

Putting it in set notation, we have

$$C = \{A \in \mathbb{S}^n \mid A - B \succeq 0\}$$

Now we know that a set is convex if

$$A_1, A_2 \in C, \quad 0 \leq \theta \leq 1 \quad \implies \quad \theta A_1 + (1 - \theta)A_2 \in C$$

Writing out the set with some $A_1, A_2 \in C$, we have

$$\theta A_1 + (1 - \theta)A_2 - B \succeq 0$$

Distributing the subtraction of B , we have

$$\theta(A_1 - B) + (1 - \theta)(A_2 - B) \succeq 0$$

Since $A_1, A_2 \in C$, we know that $A_1 - B \succeq 0$ and $A_2 - B \succeq 0$. Since this is a linear combination of PSD's, it is also PSD. We have now seen that $\theta A_1 + (1 - \theta)A_2 \in C$, so the set is convex.

Problem 5

A set is defined as

$$C = \{x \in \mathbb{R}^n \mid \|x - a\|_2 \leq \theta \|x - y\|_2 \text{ for all } y \in S\}$$

where $S \subseteq \mathbb{R}^n$.

- Determine whether the set is convex, which may depend on the value of θ . For the case that it is, provide your reasoning; otherwise, provide a counter-example.
- Let $a = (1, 0)^T$ and $S = \{(-1, 0)^T, (0, -1)^T\}$. For $\theta = 2$ and $\theta = 0.5$, sketch the resultant set, respectively.

Solution

Part (a): I will define two arbitrary points in C , now $x_1, x_2 \in C$ and check for convexity. I'll use $\phi x_1 + (1 - \phi)x_2$ to avoid confusion with θ .

$$\|\phi x_1 + (1 - \phi)x_2 - a\|_2 \leq \theta \|\phi x_1 + (1 - \phi)x_2 - y\|_2$$

Which equals

$$\|\phi(x_1 - a) + (1 - \phi)(x_2 - a)\|_2 \leq \theta \|\phi(x_1 - y) + (1 - \phi)(x_2 - y)\|_2$$

The value of θ determines the size and shape of the set.

Part (b): For $\theta = 0.5$, the set is defined as

$$\{x \in \mathbb{R}^n \mid \|x - (1, 0)^T\|_2 \leq 0.5 \|x - y\|_2 \text{ for all } y \in S\}$$

Which simplifies to the regions $(x_1 - 1)^2 + x_2^2 \leq 0.25((x_1 + 1)^2 + x_2^2)$ and $(x_1 - 1)^2 + x_2^2 \leq 0.25(x_1^2 + (x_2 + 1)^2)$. Plotting those regions

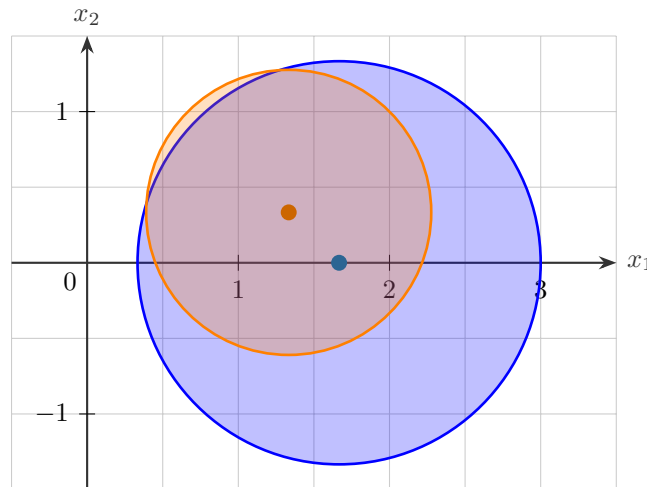


Figure 1: Set for $\theta = 0.5$

For $\theta = 2$, the set is defined as

$$\{x \in \mathbb{R}^n \mid \|x - (1, 0)^T\|_2 \leq 2 \|x - y\|_2 \text{ for all } y \in S\}$$

Which simplifies to the regions $(x_1 - 1)^2 + x_2^2 \leq 4((x_1 + 1)^2 + x_2^2)$ and $(x_1 - 1)^2 + x_2^2 \leq 4(x_1^2 + (x_2 + 1)^2)$.

Plotting those regions

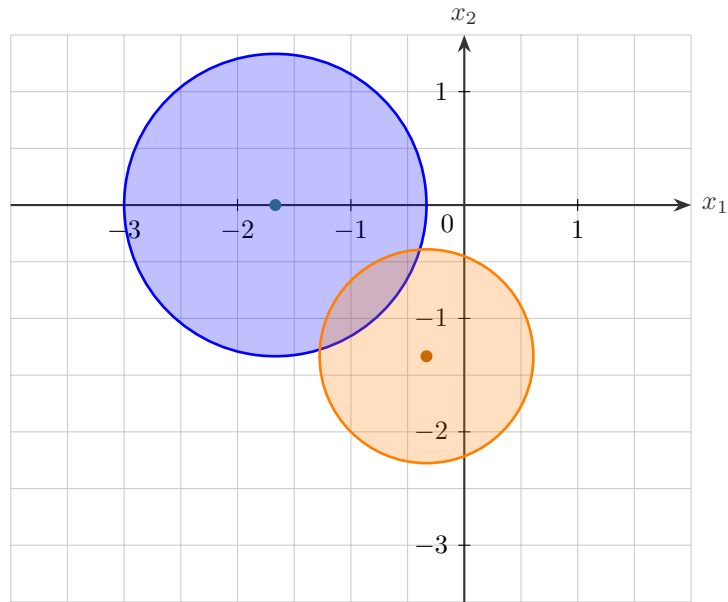


Figure 2: Set for $\theta = 2$

Problem 6

(a) Show that if S_1 and S_2 are convex set in $\mathbb{R}^n$, then the set sum

$$S \triangleq \{x_1 + x_2 \mid x_1 \in S_1, x_2 \in S_2\}$$

is also convex.

(b) Let $S_1 = \{(-1, -1)^T\}$ and $S_2 = \{(2, 2)^T + u \mid \|u\| \leq 1\}$. What is their set sum?

(c) Show that if S_1 and S_2 are convex set in $\mathbb{R}^n$, then the inverse sum

$$S \triangleq \bigcup_{1 \geq \lambda \geq 0} \{\lambda S_1 \cap (1 - \lambda) S_2\}$$

is also convex, where $\lambda S_j = \{\lambda x_j \mid x_j \in S_j\}$, $j = 1, 2$.

(d) Let $S_1 = \{(1, 1)^T\}$ and $S_2 = \{(2, 2)^T + u \mid \|u\| \leq 1\}$. What is their inverse sum?

Solution

Part (a): Similar to above, we choose two arbitray points in S

$$a = x_1 + x_2, \quad b = y_1 + y_2, \quad x_1, y_1 \in S_1, \quad x_2, y_2 \in S_2$$

and check if they satisfy the conditions of a convex set

$$\begin{aligned} \theta a + (1 - \theta)b &= \theta(x_1 + x_2) + (1 - \theta)(y_1 + y_2) \\ &= \underbrace{\theta x_1 + (1 - \theta)y_1}_{\in S_1} + \underbrace{\theta x_2 + (1 - \theta)y_2}_{\in S_2} \end{aligned}$$

So this is a sum of terms $x_1 + x_2$ where $x_1 \in S_1$ and $x_2 \in S_2$, so $\theta a + (1 - \theta)b \in S$. Thus, S is a convex set.

Part (b): The set sum of S_1 and S_2 is

$$\begin{aligned} S_1 + S_2 &= \{(-1, -1)^T + (2, 2)^T + u \mid \|u\| \leq 1\} \\ &= \{(1, 1)^T + u \mid \|u\| \leq 1\} \end{aligned}$$

Which is a circle of radius 1 centered at $(1, 1)$.

Part (c): The linear combination of two points in S is also in S , so the inverse sum is a convex set.

Part (d): The inverse sum of S_1 and S_2 is

$$S = \bigcup_{1 \geq \lambda \geq 0} \{\lambda(1, 1)^T \cap (1 - \lambda)((2, 2)^T + u) \mid \|u\| \leq 1\}$$

Problem 7

It is known that one halfspace contains another in the following form

$$\{x \mid a^T x \leq b\} \subseteq \{x \mid c \cdot a^T x \leq \tilde{b}\}.$$

Provide the conditions that the parameters c , b , and $\tilde{b}$ must satisfy. Are these conditions also sufficient for the containment relation between these two halfspaces?

Solution

For the first set to be a subset of the second, any x that satisfies $a^T x \leq b$ must also satisfy $c \cdot a^T x \leq \tilde{b}$.

If $c < 0$, then x must satisfy

$$\begin{cases} a^T x \leq b \\ c \cdot a^T x \leq \tilde{b} \end{cases} \implies \begin{cases} a^T x \leq b \\ a^T x \geq \tilde{b}/c \end{cases}$$

Which presents a contradiction, so $c \geq 0$.

For $c = 0$, we have

$$\begin{cases} a^T x \leq b \\ 0 \leq \tilde{b} \end{cases}$$

Which is valid when $\tilde{b} \geq 0$.

For $c > 0$

$$\begin{cases} a^T x \leq b \\ c \cdot a^T x \leq \tilde{b} \end{cases}$$

Which requires that $b \leq \tilde{b}/c$.

Problem 8

For a pair of given sets $C, D \subseteq \mathbb{R}^n$, consider the following set

$$S = \{(a, b) \in \mathbb{R}^{n+1} \mid a^T x \leq b \text{ for all } x \in C, a^T x \geq b \text{ for all } x \in D\}.$$

Prove that the set S is convex.

Solution

Choosing two arbitrary points in S , we have

$$x = (a_1, b_1), \quad y = (a_2, b_2)$$

and applying it to our convex definition

$$\theta x + (1 - \theta)y = (\theta a_1 + (1 - \theta)a_2, \theta b_1 + (1 - \theta)b_2)$$

Now to see if this is in S , we check the two conditions, starting with the first

$$(\theta a_1 + (1 - \theta)a_2)^T x = \theta a_1^T x + (1 - \theta)a_2^T x \leq \theta b_1 + (1 - \theta)b_2, \quad \forall x \in C$$

and then the second

$$(\theta a_1 + (1 - \theta)a_2)^T x = \theta a_1^T x + (1 - \theta)a_2^T x \geq \theta b_1 + (1 - \theta)b_2, \quad \forall x \in D$$

Since both of these conditions hold by the set definition, the set S is convex.

Problem 9

Let $C = \{X \in \mathbb{R}^{n \times n} \mid a^T X a \geq 0, \forall a \in \mathbb{R}^n\}$. Determine whether the set

$$\mathcal{S} = \{A \in \mathbb{R}^{n \times n} \mid A - M \in C\}$$

is convex, where $M \in C$ is a fixed matrix. Provide justification or a counterexample. Note that usually the definition of a matrix being positive-definite requires the matrix to be symmetric, but this problem removes that condition.

Solution

Again choose two arbitrary points in $\mathcal{S}$

$$A_1, A_2 \in \mathcal{S}$$

and check the convex condition

$$\theta A_1 + (1 - \theta) A_2 - M \in C$$

Distributing the subtraction of M , we have

$$\theta(A_1 - M) + (1 - \theta)(A_2 - M) \in C$$

Now this new matrix is in C if

$$a^T(\theta(A_1 - M) + (1 - \theta)(A_2 - M))a \geq 0$$

Which becomes

$$a^T(\theta(A_1 - M))a + a^T((1 - \theta)(A_2 - M))a \geq 0$$

Since $A_1, A_2 \in \mathcal{S}$, we know that $A_1 - M \in C$ and $A_2 - M \in C$, so these terms are ≥ 0 . Set $\mathcal{S}$ satisfies the convex conditions, so it is convex.

Problem 10

Plot the norm balls of radius $r = 1$ for the following norm in $\mathbb{R}^2$ and $\mathbb{R}^3$:

- (a) ℓ_1 norm;
- (b) ℓ_∞ norm;
- (c) ℓ_2 norm.

Solution

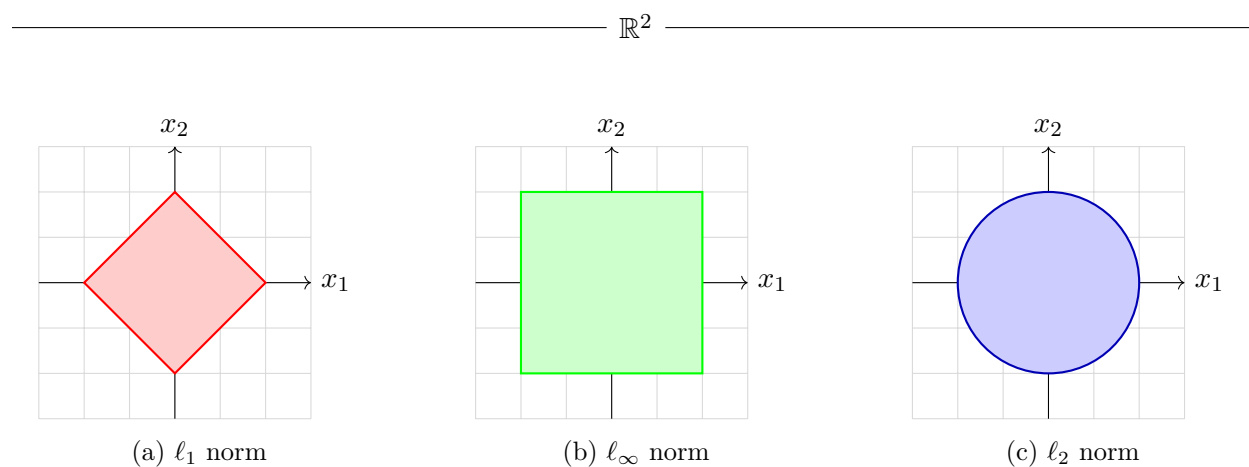


Figure 3: Norm balls in $\mathbb{R}^2$

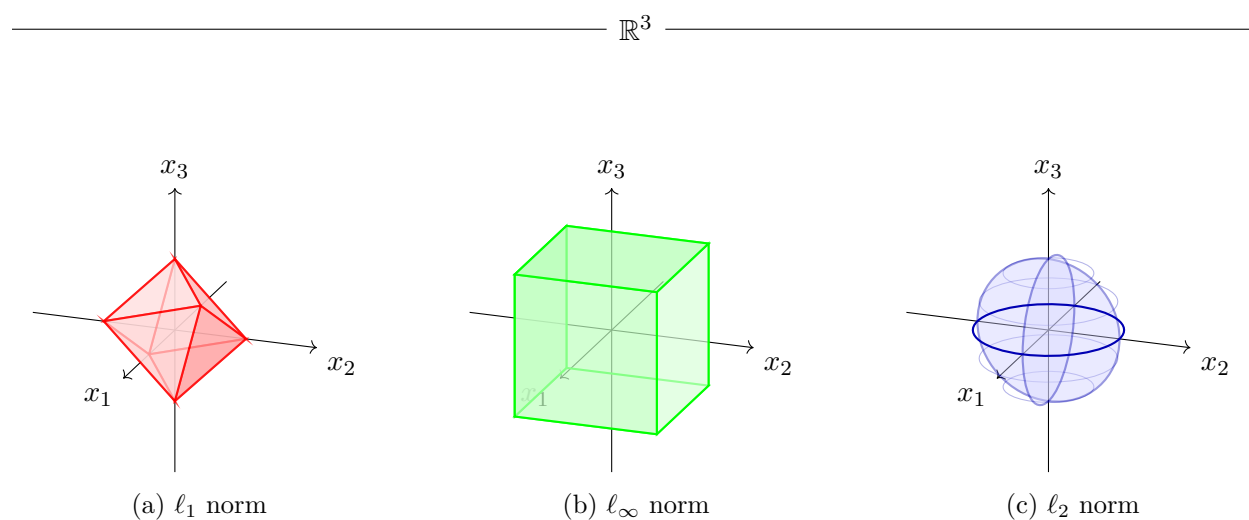


Figure 4: Norm balls in $\mathbb{R}^3$